Toward a Zero-Trust Architecture for 6G-Enabled E-Business Ecosystem *

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Abstract

E-Business ecosystems, encompassing digital payments, e-commerce, logistics, and B2B platforms, face growing security challenges as they evolve toward 6G-enabled ultra-connected environments. Heterogeneous actors and sensitive data flows render traditional perimeter-based models inadequate, requiring continuous verification and adaptive controls. Zero-Trust Architecture (ZTA), as defined in NIST SP 800-207, provides a promising paradigm but demands domain-specific adaptation. This paper bridges the gap between standardization-driven research and practical security needs. We first present a "Survey of Surveys" to highlight the lack of holistic ZT integration in prior studies. Subsequently, we evaluate the applicability of ZTA to 6G E-Business by leveraging the key issues and thirteen candidate solutions outlined in 3GPP TR 33.794. We identify which solutions are most suitable for transactional integrity, API security, and dynamic enforcement in digital commerce, offering tailored guidance for building secure and trustworthy digital ecosystems.

1 Introduction

The sixth-generation (6G) network is envisioned not merely as an extension of 5G connectivity, but as a hyper-connected platform integrating physical and digital worlds, often referred to as the "Internet of Skills" or "Ubiquitous Intelligence" [6, 16]. Unlike previous generations focused on data throughput, 6G is expected to serve as a critical infrastructure for E-Business ecosystems, supporting autonomous economic transactions, real-time micro-payments, and mission-critical industrial applications [5, 14]. As global standardization bodies like ITU-R and 3GPP define the framework for IMT-2030, the integration of non-terrestrial networks (NTN) and AI-native architectures implies that 6G will handle unprecedented levels of sensitive financial and personal data [11].

However, the security paradigms inherited from 5G are increasingly insufficient for this new landscape. While 5G introduced robust security features such as Service Based Architecture (SBA) security and unified authentication [2, 1], it essentially relies on a perimeter-based defense model. A critical vulnerability in current 5G networks is the "Implicit Trust" assumption within the Radio Access Network (RAN) and backhaul infrastructure; once a network entity (e.g., gNB or Femto cell) is authenticated, it is often trusted for the duration of its operation without continuous re-verification [12].

Recent security incidents demonstrate that this implicit trust is a fatal flaw in E-Business scenarios. Sophisticated attacks are shifting from external jamming to compromising internal trusted nodes. Research indicates that compromised edge nodes or rogue base stations can

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manipulate user plane traffic to intercept financial transactions or launch Man-in-the-Middle (MitM) attacks on E-Business services [7, 8]. In a 6G environment where machines conduct autonomous payments, such breaches could lead to direct, irreversible financial losses rather than simple service disruptions. Thus, the traditional binary trust model—"Trusted Internal" vs. "Untrusted External"—is no longer viable.

To address these challenges, Zero Trust Architecture (ZTA), grounded in the principle of "Never Trust, Always Verify," has emerged as a necessary paradigm [13]. 3GPP has recognized this necessity, recently initiating studies such as TR 33.794 to enable ZTA within the 5G Core [4, 3]. However, current research and standardization efforts predominantly focus on the Core Network (SBA) and cloud-native interfaces, largely overlooking the practical implementation of ZT in the RAN and user-plane data paths. Most existing surveys either treat ZT as a generic feature or fail to address the specific requirements of E-Business continuity in mobile networks.

In this paper, we bridge this gap by analyzing the limitations of existing security surveys and defining a 6G-Native Zero Trust Architecture tailored for E-Business ecosystems. Unlike previous works that isolate ZT within the Core, we structure an End-to-End ZT model that extends continuous verification to the RAN edge. Our contribution is twofold: first, we provide a "Survey of Surveys," critically evaluating prior 5G/6G security reviews to highlight the lack of holistic ZT integration. Second, we derive architectural requirements to mitigate internal threats in 6G financial services, moving beyond conceptual discussions to practical realization.

2 Related Work: A Survey of Surveys

As the concept of Zero Trust gains traction in telecommunications, numerous survey papers have attempted to analyze its applicability to 5G and 6G networks. However, most existing studies focus either on general security landscapes or strictly on Core Network architectural changes, often overlooking the specific requirements of vertical domains like E-Business. This section classifies existing literature and highlights the research gap this paper aims to fill.

2.1 General 5G/6G Security Surveys vs. 6G-Native Zero Trust Architecture

Existing literature on next-generation network security can be broadly categorized into two streams, both of which fall short of addressing the specific needs of E-Business ecosystems.

First, **General 5G/6G Security Surveys** [8, 12] provide comprehensive overviews of the threat landscape, covering physical layer jamming, AI-enabled attacks, and privacy-preserving techniques. While these studies mention Zero Trust (ZT) as a potential countermeasure, they treat it merely as a conceptual feature or a buzzword, lacking a structural analysis of how ZT principles should be enforced across the heterogenous Radio Access Network (RAN) and Edge environments [11].

Second, **SBA-Centric ZT Studies** [7, 3] focus on securing the Core Network by leveraging the Service Based Architecture (SBA). Although they effectively address cloud-native threats (e.g., container security, service mesh), they inherently ignore the "Implicit Trust" vulnerability in the RAN. This omission is critical for E-Business, as financial transactions often originate from mobile devices vulnerable to Rogue Base Station attacks, which purely Core-centric models cannot detect [9].

Table 1 contrasts our work with these existing studies, highlighting the necessity of an "End-to-End Trust" model that integrates RAN-level context with business logic.

Table 1: Comparison of Related Surveys and Research Gap

Category	Focus & Scope	Limitations (Research Gap)
General 5G/6G	• Broad threat landscape • AI/ML security & Privacy	• Treats ZT as a conceptual feature • Lacks architectural implementation details
SBA-Centric ZT	• **Core Network** focus (3GPP SBA) • Cloud-native security	• **Ignores RAN risks** (e.g., Rogue gNB) • Insufficient for E-Business transaction integrity
This Work	• **E-Business Ecosystem** in 6G • **End-to-End ZT** (RAN + Core + App)	• **Addresses the Gaps**: Defines practical architectural requirements for secure digital commerce.

2.2 The Gap between Enterprise ZT and Telco ZT

Another critical limitation in current research is the direct application of Enterprise IT Zero Trust models to mobile networks without adaptation [13, 10]. As illustrated in Figure 1, the standard Zero Trust Architecture (ZTA) defined in NIST SP 800-207 operates on a logical framework where the Policy Decision Point (PDP) and Policy Enforcement Point (PEP) govern access based on relatively static attributes [13]. While this centralized model is effective for securing corporate assets in fixed environments, it fails to account for the extreme performance and mobility requirements of next-generation mobile networks [6, 15].

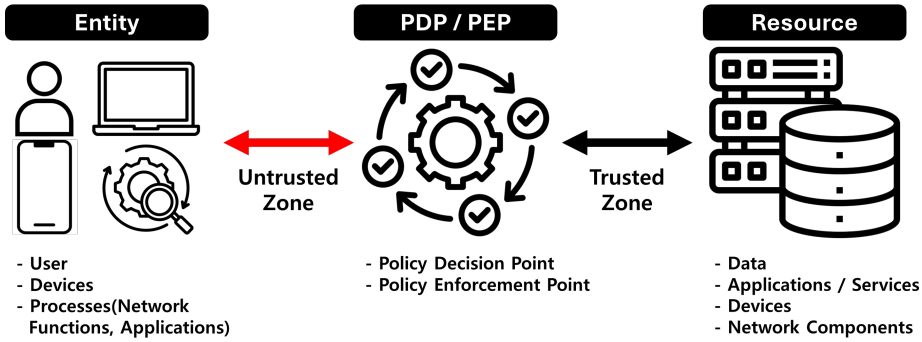


Figure 1: Logical components of NIST SP 800-207 Zero Trust Architecture [13].

In contrast, the 6G-enabled E-Business ecosystem is characterized by high mobility, ultra-low latency, and massive heterogeneity [5]. As 6G vision involves dynamic actors such as V2X, autonomous UAVs, and XR devices, the security framework must ensure seamless Handover integrity and continuous verification without introducing significant signaling overhead that could violate URLLC constraints [11]. Unlike the identity-centric focus of traditional models, a "6G-Native Zero Trust" must incorporate network-specific contexts including network slicing, real-time location, and RAN conditions to protect autonomous financial transactions [12].

As summarized in Table 2, this paper differentiates itself by addressing these unique constraints, moving beyond standard IT-centric studies to ensure the infrastructure integrity re-

quired for modern digital commerce.

Table 2: Gap Analysis: Enterprise ZT vs. 6G-Native ZT Requirements

Feature	**Enterprise ZT** (Traditional)	**6G-Native ZT** (Required)
Environment	Static (offices, laptops)	**Dynamic & Mobile** (V2X, IoT, XR)
Performance	Human-speed latency	**Ultra-Low Latency (URLLC)**
Trust Scope	Identity & Device Health	**Slice, Location & RAN Condition**
Challenge	Internal Lateral Movement	**Handover & Infrastructure Integrity**

2.3 Critical Limitations of Current Approaches

Despite the extensive literature on 5G security, two fundamental gaps remain unaddressed for E-Business adoption.

First, the *Silo Effect between RAN and Core* is a major barrier. Most ZT studies focus exclusively on the Service Based Architecture (SBA) in the Core Network, assuming the RAN is merely a transparent data pipe. However, in 6G E-Business scenarios, the RAN Edge is often the first point of contact for financial transactions. Existing perimeter-based models fail to extend the verification phase to the physical link level, leaving the system vulnerable to physical layer attacks such as Rogue Base Stations.

Second, the *Disconnect between Network Security and Business Logic* limits practical application. Current 3GPP standards (e.g., TS 33.501) provide robust connectivity security but lack semantic awareness of the application layer. For instance, a "valid" network connection does not imply a "legitimate" financial transaction. There is a lack of architectural frameworks that correlate network telemetry (e.g., latency, location) with business-level anomalies (e.g., fraud patterns), which is the primary focus of this paper.

3 6G-Native Zero Trust Architecture for E-Business

Instead of treating Zero Trust (ZT) as a supplementary feature, we define an operational architecture that embeds ZT principles directly into the 6G E-Business transaction flow. This section details how we operationalize the candidate solutions from 3GPP TR 33.794 [4] to bridge the gap identified in Section 2.

3.1 Architectural Overview

The defined model shifts the trust boundary from the "Network Edge" to the "Transaction Context." Unlike the traditional 5G security model which relies on static, perimeter-based trust boundaries (e.g., standard AKA and IPsec), our architecture enforces dynamic, per-transaction Continuous Verification. We clarify that standard 5G security features are foundational enablers, but they do not constitute a Zero Trust Architecture by themselves without the continuous decision loops proposed herein.

To realize this, we enforce verification at three distinct layers:

1. **Radio Layer (RAN):** Verifying physical attributes (location, beam ID) to detect Rogue Base Stations.

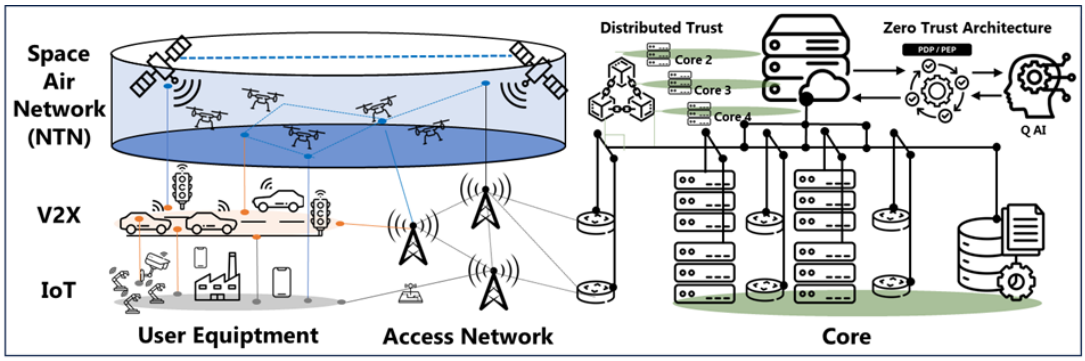


Figure 2: Schematic of 6G-Native Zero Trust Architecture with Edge-Distributed PDP/PEP. Unlike the centralized NIST model, this architecture distributes enforcement to the Radio Edge.

2. **Network Layer (SBA):** Monitoring API invocation patterns to detect token theft.
3. **Application Layer (Biz):** Validating transaction logic against user behavior profiles.

3.2 Operationalizing 3GPP Enablers for E-Business

To realize this layered defense, we select and tailor specific 3GPP enablers. It is important to note that our mapping is 'Logical', not 'Physical'. We leverage existing Network Functions (e.g., PCF, UPF) to act as Policy Decision/Enforcement Points to minimize architectural changes, but the decision logic within them is augmented with specific ZT algorithms (e.g., impossible travel detection) that are not present in standard 5G specifications.

Table 3 presents our "Operational Mapping," distinguishing between solutions essential for financial integrity and those that are redundant for E-Business.

3.3 Key Architectural Decisions

Based on the mapping in Table 3, we define two critical enforcement mechanisms for E-Business.

3.3.1 Context-Aware Token Binding (Sol#1 + Sol#12)

Traditional OAuth tokens are static. We propose binding the *Physical Cell ID (PCI)* and *User Location* to the short-lived access token (Sol#12). This ensures that a token stolen from a legitimate user cannot be replayed from a different location, effectively neutralizing the "Implicit Trust" vulnerability of the RAN.

Note that this differs from standard 5G re-authentication mechanisms, which are triggered by timer expiration or mobility events. Our approach binds verification strictly to the 'Transaction Context' (e.g., payment request), ensuring validation occurs at the exact moment of value transfer, not just during session maintenance.

3.3.2 AI-Driven Closed-Loop Response (Sol#7 + Sol#11)

Manual intervention is too slow for 6G financial speeds. We utilize Sol#7 (API Monitoring) as the trigger and Sol#11 (Closed-loop) as the actuator. If the AI engine detects an anomaly score above the threshold during a transaction request, the policy engine (PCF) executes a "Block"

Table 3: Operational Mapping: 3GPP ZT Solutions Tailored for 6G E-Business

Sol. ID	**Mechanism**	**E-Business Operationalization (Our Proposal)**	**Suitability**
Pillar 1: Visibility & Data Exposure			
Sol#1, #2	Net/Service Data Collection	We map these to **Transaction Log Exporting**. The UPF exposes packet drops or latency spikes to the Banking FDS to correlate network issues with payment failures.	**High**
Sol#7	API Anomaly Detection	Used to detect **"Impossible Travel"**. If a payment token is used from two distant gNBs simultaneously, the API gateway triggers an alarm.	**High**
Sol#8	UE Behavior Monitoring	Analyzes device patterns to distinguish legitimate users from **Botnets** attempting DDoS on payment gateways.	**High**
Pillar 2: Dynamic Enforcement			
Sol#11	Closed-loop Enforcement	Enables **Automated Quarantine**. Upon detecting a rogue gNB (via Sol#1), the PCF immediately **quarantines the specific user session** or isolates the compromised slice **instance**. This serves as a reactive containment measure complementing standard network slicing security (e.g., NSSAAF).	**High**
Sol#12	Short-lived Tokens	Implements **Micro-segmentation**. A unique cryptographic token is issued per transaction, neutralizing Replay Attacks even if traffic is intercepted.	**High**
Sol#13	NF-to-NF Trust	*Excluded*. Complex telco-internal scoring is irrelevant for external Fintech APIs.	Low

action at the edge (UPF) within milliseconds, preventing financial loss before the transaction reaches the core ledger.

4 Evaluation and Discussion

While the defined Zero-Trust Architecture offers robust security for 6G E-Business, its practical implementation introduces several challenges that must be addressed to balance security with performance.

4.1 Security vs. Ultra-Low Latency (URLLC)

One of the core promises of 6G is Ultra-Reliable Low Latency Communication (URLLC). However, the continuous verification mechanism of ZTA inevitably introduces processing overhead. For instance, Sol#11 (Closed-loop Enforcement) and Sol#12 (Short-lived Tokens) require frequent signaling between the PEP (User Plane) and PDP (Control Plane). In high-frequency trading or real-time payment scenarios, even a microsecond delay caused by token validation can be critical. Therefore, we emphasize the need for a Lightweight ZT Protocol specifically optimized for 6G hardware acceleration to ensure that security checks do not become a bottleneck

for E-Business transactions.

4.2 Privacy Concerns in Data Exposure

Sol#1 and Sol#2 rely on exposing internal network logs and user behavior data to analytics engines (NWDAF). In an E-Business context, this data often contains sensitive financial information and user location history. Exposing raw data to third-party security vendors or cloud-based PDPs raises significant privacy concerns and may violate regulations like GDPR. To mitigate this, Privacy-Preserving Analytics techniques, such as Federated Learning or Homomorphic Encryption, should be integrated into the data collection layer to perform anomaly detection without revealing raw user data.

4.3 Adversarial Attacks on AI-Driven Enforcement

The architecture heavily relies on AI/ML for anomaly detection (Sol#7, Sol#11). However, AI models themselves are vulnerable to Adversarial Attacks. Sophisticated attackers could inject carefully crafted noise into the transaction traffic to deceive the NWDAF into classifying malicious behavior as benign (Model Poisoning). For a resilient 6G E-Business ecosystem, the ZT framework must include "AI Security" measures, such as adversarial training and model robustness verification, to prevent the security engine itself from being compromised.

4.4 Qualitative Security Analysis

We evaluate the architecture against three critical threat vectors prevalent in the 6G E-Business ecosystem.

4.4.1 Defense against Credential Theft and Replay

In traditional systems, a stolen long-term session key allows an attacker to impersonate a user indefinitely until the session expires. Our architecture mitigates this via *Context-Aware Token Binding* (Sol#12). By binding the User Equipment's (UE) real-time physical cell ID and beamforming signature to the transaction token, any attempt to replay the token from a different location or device results in immediate verification failure at the PEP, even if the cryptographic signature is valid.

4.4.2 Mitigation of Insider Threats

A compromised internal node (e.g., a hijacked edge server) poses a severe risk in the "Implicit Trust" model. By enforcing Sol#7 (API Anomaly Detection), our model monitors the behavioral patterns of internal Network Functions. If a trusted NF begins making irregular high-frequency queries to the Banking API—indicative of data exfiltration—the NWDAF identifies this deviation from the baseline profile. Subsequently, Sol#11 triggers an automated policy update to isolate the compromised NF from the slice, limiting the blast radius of the attack.

4.4.3 Resilience against DDoS on Payment Gateways

6G massive IoT environments increase the risk of botnet-driven DDoS attacks. Traditional firewalls often fail to distinguish legitimate flash crowds from malicious floods. Our approach utilizes Sol#8 (UE Behavior Monitoring) to analyze the signaling patterns of devices at the

RAN level. By distinguishing human transaction intervals from automated botnet frequencies, the system can preemptively throttle malicious traffic at the Edge UPF before it saturates the central payment gateways.

5 Conclusion

This paper examined how the Zero-Trust Architecture (ZTA) defined in NIST SP 800-207 and its enablers proposed in 3GPP TR 33.794 can be adapted to 6G-enabled E-Business ecosystems. While the 3GPP report originally targets network-level trust management for operator domains, our analysis identified which of its thirteen candidate solutions can be effectively mapped to business-centric, API-driven environments.

By aligning the key issues—data exposure for monitoring and dynamic policy enforcement—with the specific requirements of digital commerce, this work establishes a framework for interpreting standardized ZTA mechanisms through an E-Business perspective. The results show that not all solutions are equally applicable to cross-domain business ecosystems. Data-centric approaches such as network-assisted data collection, security log correlation, and API risk monitoring align naturally with the telemetry and fraud analytics already present in modern digital platforms. Likewise, policy-centric mechanisms such as adaptive access control, AI-based enforcement, and short-lived credentials complement the dynamic, high-value transactions of E-Business without requiring telecom-specific infrastructure.

These findings confirm that the Zero-Trust paradigm can be realized in 6G business ecosystems through lightweight, context-aware implementations rather than rigid NF extensions. Future research should focus on operationalizing these insights within standardized frameworks, particularly the integration of AI-native PDP/PEP modules capable of federated learning across financial, logistics, and customer domains. Ultimately, the convergence of ZTA principles, AI-native enforcement, and quantum-safe cryptography will form a unified foundation for secure and trustworthy 6G E-Business ecosystems.

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